

DIGITAL LOGIC DESIGN (EE1005) - MASTER FINAL EXAM REVISION GUIDE

FAST NUCES | Exam Weightage: Mid1 (15%) + Mid2 (20-25%) + Post-Mid2 (60-65%)

CONTENT DISTRIBUTION BY WEIGHTAGE

Period	Weightage	Topics	Pages
Mid 1	15%	Number Systems, Boolean Algebra, K-Maps, Gates	§1-2
Mid 2	20-25%	Combinational Logic (MUX, Decoder), Adders	§3-4
Post-Mid 2	60-65%	Sequential Logic, Latches, Flip-Flops, Counters, Shift Registers	§5-8

SECTION 1: MID 1 TOPICS (15% Weightage)

1.1 NUMBER SYSTEMS & CONVERSIONS

Binary-Decimal Conversions

- **Decimal to Binary:** Divide by 2 repeatedly, collect remainders
- **Binary to Decimal:** Sum of (bit × 2^{position})
- **Hex to Binary:** Each hex digit = 4 binary bits
- **Gray Code:** MSB same as binary MSB; subsequent bits = XOR of previous binary bits
 - Gray code advantage: Only 1 bit changes per transition (eliminates race conditions)

Signed Number Representation

Method	Range (4-bit)	Example (-5)	Key Use
Sign-Magnitude	-7 to +7	1101	Historical
1's Complement	-7 to +7	1010	Rarely used
2's Complement	-8 to +7	1011	Standard

2's Complement Calculation: Flip all bits, add 1.

1.2 BOOLEAN ALGEBRA & LOGIC GATES

Core Theorems (DeMorgan's Law Critical)

- **DeMorgan's 1st:** $(A \cdot B)' = A' + B'$
- **DeMorgan's 2nd:** $(A + B)' = A' \cdot B'$
- **Involution:** $(A')' = A$
- **Absorption:** $A + A \cdot B = A$; $A \cdot (A + B) = A$
- **Distributive:** $A \cdot (B + C) = A \cdot B + A \cdot C$

Truth Tables for Critical Gates

Input	NOT	AND	OR	XOR	NAND	NOR	XNOR
0,0	—	0	0	0	1	1	1
0,1	—	0	1	1	1	0	0
1,0	—	0	1	1	1	0	0
1,1	—	1	1	0	0	0	1

Universal Gates: NAND, NOR can implement any logic.

1.3 KARNAUGH MAP (K-MAP) - ESSENTIAL FOR 10-15 MARKS

K-Map Minimization Rules (HIGH YIELD)

1. **Group adjacent 1s** in powers of 2 (2, 4, 8, 16 cells)
2. **Largest groups first** → fewer literals in final expression
3. **Don't cares (X)** → include if they reduce group count
4. **Wrap-around allowed** → edges and corners are adjacent
5. **Output 0:** Group 0s for complemented form, then apply DeMorgan's

4-Variable K-Map Layout (Most Common in Exams)

	AB			
CD	00	01	11	10
00	0	1	2	3
01	4	5	6	7
11	12	13	14	15
10	8	9	10	11

Adjacency Rule: Row labels must differ by 1 bit; column labels must differ by 1 bit.

K-Map Pattern Recognition (Exam Strategy)

- **Single 1 in corner:** Must be included in pair or quad to minimize
 - **Checkerboard pattern:** Cannot simplify beyond individual minterms
 - **All 1s:** Output = 1 (regardless of inputs)
 - **All 0s:** Output = 0
-

1.4 COMBINATIONAL CIRCUIT DESIGN METHODOLOGY

Step-by-Step Process

1. **Write truth table** with inputs and outputs
2. **Extract minterms** (rows where output = 1)
3. **Draw K-Map** and group 1s
4. **Read simplified expression** from groups
5. **Implement using gates** (AND-OR, NAND, or NOR)
6. **Verify** against original truth table

Example: 3-Input Majority Function (Exam Pattern)

- Output = 1 if at least 2 inputs = 1
 - Truth table: m(3,5,6,7)
 - K-Map groups: Two 4-cell groups ($AB + AC + BC$)
 - Circuit: 3 AND gates $\rightarrow$ 1 OR gate
-
-

SECTION 2: MID 2 TOPICS (20-25% Weightage)

2.1 MULTIPLEXERS (MUX) & DECODERS

4-to-1 Multiplexer (Basic Building Block)

Truth Table: Select $S[1:0]$ determines output

S_1	S_0	Output Y
-------	-------	------------

0	0	I_0
---	---	-------

0	1	I_1
---	---	-------

1	0	I_2
---	---	-------

1	1	I_3
---	---	-------

Logic: $Y = S_1' \cdot S_0' \cdot I_0 + S_1' \cdot S_0 \cdot I_1 + S_1 \cdot S_0' \cdot I_2 + S_1 \cdot S_0 \cdot I_3$

Key Property: MUX is programmable logic → can implement any function with sufficient inputs.

Decoder (3-to-8 Binary Decoder - Exam Standard)

Inputs: A2 A1 A0 (3-bit binary code)

Outputs: Y0 to Y7 (one-hot encoding)

Y0 = A2' · A1' · A0' (active when input = 000)

Y1 = A2' · A1' · A0 (active when input = 001)

...

Y7 = A2 · A1 · A0 (active when input = 111)

Only ONE output = 1 at any time

Critical Distinction

Feature	MUX	Decoder
Function	Select 1 of N inputs	Activate 1 of N outputs
Inputs	Data lines + Select lines	Data only
Outputs	1 output	N outputs (one-hot)
Use Case	Data routing	Address decoding

2.2 ADDERS & ARITHMETIC CIRCUITS

Half Adder (HA)

Inputs: A, B

Outputs: Sum (S), Carry (C)

S = A XOR B

C = A AND B

Truth Table:

A B | S C

0 0 | 0 0

0 1 | 1 0

1 0 | 1 0

1 1 | 0 1

Full Adder (FA) - HIGH YIELD

Inputs: A, B, Cin (carry-in)

Outputs: Sum (S), Cout (carry-out)

$$S = A \oplus B \oplus C_{in}$$

$$\begin{aligned} C_{out} &= A \cdot B + C_{in} \cdot (A \oplus B) \\ &= A \cdot B + C_{in} \cdot A + C_{in} \cdot B \end{aligned}$$

Truth Table (8 rows - memorize patterns):

A	B	Cin	S	Cout
---	---	-----	---	------

0	0	0	0	0
---	---	---	---	---

0	0	1	1	0
---	---	---	---	---

0	1	0	1	0
---	---	---	---	---

0	1	1	0	1
---	---	---	---	---

1	0	0	1	0
---	---	---	---	---

1	0	1	0	1
---	---	---	---	---

1	1	0	0	1
---	---	---	---	---

1	1	1	1	1
---	---	---	---	---

Pattern: S = 1 when odd number of inputs = 1

Cout = 1 when 2+ inputs = 1

4-Bit Ripple Carry Adder (RCA)

- Cascade 4 Full Adders
 - Cin of each stage = Cout of previous stage
 - **Disadvantage:** Propagation delay increases with bit width
 - **Advantage:** Simple, minimal gates
-
-

SECTION 3: EARLY POST-MID 2 - LATCHES & FLIP-FLOPS (25-30%)

3.1 S-R LATCH (NOR-Based)

Truth Table & Behavior

S	R	Q	Q'	Behavior
0	0	Q	Q'	Hold (no change)
0	1	0	1	Reset (Q = 0)
1	0	1	0	Set (Q = 1)
1	1	?	?	INVALID (race condition)

Circuit: Two cross-coupled NOR gates

Critical Point: S=1, R=1 produces undefined state (metastability).

3.2 D LATCH (Transparent Latch)

Clocking Mechanism

When Enable (E) = 1:

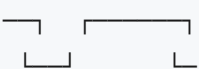
Q follows D immediately (transparent)

When Enable (E) = 0:

Q holds previous value

Timing Diagram:

Clock  (period shown)

D 

Q  (follows D with propagation delay)

Key Issue: Level-triggered → vulnerable to setup/hold time violations.

3.3 D FLIP-FLOP (Edge-Triggered)

Master-Slave Configuration

Master latch (level-triggered)

↓ (enabled during inactive clock)

Slave latch (level-triggered)

↓ (enabled during active clock)

Result: Edge-triggered behavior

On rising edge of clock:

1. Master captures D

2. Slave updates Q

3. Q changes at clock edge (synchronous)

Timing Parameters (CRITICAL FOR DESIGN)

Parameter	Symbol	Definition	Impact
Setup Time	tsu	Min. time D stable BEFORE clock edge	Must satisfy to prevent metastability
Hold Time	th	Min. time D stable AFTER clock edge	Prevents data corruption
Propagation Delay	tpd	Time from clock edge to Q stabilization	Limits maximum clock frequency

Design Rule: Data must be stable [tsu before Clk, th after Clk].

Timing Violation Consequences

- **Setup violation:** D changes within tsu before clock → metastability → unpredictable Q
 - **Hold violation:** D changes within th after clock → previous data lost → unpredictable Q
-

3.4 J-K FLIP-FLOP (Functional Enhancement)

Truth Table & Behavior

J	K	Q(next)	Behavior
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	Q'	Toggle

Circuit: MUX controls D input:

$$D = J \cdot Q' + K' \cdot Q$$

Key Advantage: J=K=1 produces toggle (J-K avoids invalid state of S-R).

Timing Diagram Example (4-Clock Cycle)

Clock:

J:

K:

Q (0=0):

After Clk1: J=1,K=0 → Q=1

After Clk2: J=1,K=0 → Q=1 (hold)

After Clk3: J=0,K=1 → Q=0

After Clk4: J=0,K=1 → Q=0 (hold)

3.5 T FLIP-FLOP (Toggle)

Definition & Truth Table

T | Q(next)

0 | Q (no change)

1 | Q' (toggle)

Implementation: $J=K=T$, or use D flip-flop with $D = T \cdot Q' + T' \cdot Q = T \text{ XOR } Q$

Application: Frequency divider, binary counter.

SECTION 4: SEQUENTIAL CIRCUITS - SYNCHRONOUS COUNTERS (40% Weightage - HIGHEST PRIORITY)

4.1 COUNTER FUNDAMENTALS

Classification

Counters

├ Asynchronous (Ripple)

| └ Advantage: Simple circuits

| └ Disadvantage: Propagation delays, race conditions

└ Synchronous

├ Advantage: All FF update simultaneously, no race conditions

└ Disadvantage: Complex logic

Synchronous Counter Design Methodology (EXAM-CRITICAL)

5-Step Process:

1. Construct State Transition Table

- Current state ($Q_2Q_1Q_0$) → Next state ($Q_2^+Q_1^+Q_0^+$)
- Include all N states for an N-state counter
- Unused states → Don't care (X) if 2^n states exceed count

2. Determine J-K Inputs for Each Flip-Flop

- For each FF, find J,K logic from state transitions:

$Q \rightarrow Q^+$		J	K	
$0 \rightarrow 0$		0	X	(no change)
$0 \rightarrow 1$		1	X	(set)
$1 \rightarrow 0$		X	1	(reset)
$1 \rightarrow 1$		X	0	(hold)

1. Create K-Maps for J & K Inputs

- Treat each state bit as variable
- Group 1s and X's to minimize logic
- Separate K-maps for each J-K pair

2. Read Minimized Equations from K-Maps

- Example: $J_0 = Q_2 \cdot Q_1$, $K_0 = Q_2^+ Q_1$

3. Implement Circuit

- Use AND/OR gates and J-K FF with clock

4.2 STANDARD COUNTERS

3-Bit Binary Up Counter (0→1→2→3→4→5→6→7→0)

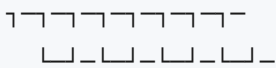
State Table:

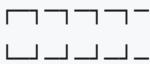
Q_2	Q_1	Q_0		Q_2^+	Q_1^+	Q_0^+		J_2	K_2		J_1	K_1		J_0	K_0
0	0	0		0	0	1		0	X		0	X		1	X
0	0	1		0	1	0		0	X		1	X		X	1
0	1	0		0	1	1		0	X		X	0		1	X
0	1	1		1	0	0		1	X		X	1		X	1
1	0	0		1	0	1		X	0		0	X		1	X
1	0	1		1	1	0		X	0		1	X		X	1
1	1	0		1	1	1		X	0		X	0		1	X
1	1	1		0	0	0		X	1		X	1		X	1

Minimized Equations (from K-maps with don't-cares):

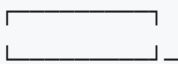
$J_0 = 1$, $K_0 = 1$ (toggle every clock)
 $J_1 = Q_0$, $K_1 = Q_0$ (Q_0 is enable)
 $J_2 = Q_1 \cdot Q_0$, $K_2 = Q_1 \cdot Q_0$ (enable when lower bits = 11)

Timing Diagram (4 clock cycles):

Clock: 

Q0: 

Q1: 

Q2: 

Count: 0 1 2 3 4 5 6 7 0

3-Bit Binary Down Counter (7→6→5→4→3→2→1→0→7)

Logic Change:

$J_1 = Q_0'$, $K_1 = Q_0'$ (enable when $Q_0 = 0$)

$J_2 = Q_1' \cdot Q_0'$, $K_2 = Q_1' \cdot Q_0'$ (enable when lower bits = 00)

4.3 CUSTOM SEQUENCE COUNTERS (HIGHEST EXAM PRIORITY)

Problem Type Analysis: Non-Binary Sequences

Frequency of Appearance: 100% of papers (average 6-8 marks) **Difficulty:** ★★★★★

Problem 1: 8-State Non-Binary Counter (2025 Exam)

Sequence: 0 → 1 → 3 → 2 → 6 → 7 → 5 → 4 → (repeat)

Solution Steps:

1. State Transition Table:

State	Binary	Next State	Next Binary	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+
0	000	1	001	0	0	0	0	0	1
1	001	3	011	0	0	1	0	1	1
3	011	2	010	0	1	1	0	1	0
2	010	6	110	0	1	0	1	1	0
6	110	7	111	1	1	0	1	1	1
7	111	5	101	1	1	1	1	0	1
5	101	4	100	1	0	1	1	0	0
4	100	0	000	1	0	0	0	0	0

1. Extract J-K Requirements:

From state 000 $\rightarrow$ 001:

- Q_2 : $0 \rightarrow 0$, $J_2=0$, $K_2=X$
- Q_1 : $0 \rightarrow 0$, $J_1=0$, $K_1=X$
- Q_0 : $0 \rightarrow 1$, $J_0=1$, $K_0=X$

From state 001 $\rightarrow$ 011:

- Q_2 : $0 \rightarrow 0$, $J_2=0$, $K_2=X$
- Q_1 : $0 \rightarrow 1$, $J_1=1$, $K_1=X$
- Q_0 : $1 \rightarrow 1$, $J_0=X$, $K_0=0$

(Continue for all 8 transitions)

1. Build K-Maps for J_2 , K_2 , J_1 , K_1 , J_0 , K_0

K-Map for J_2 (when does Q_2 go from $0 \rightarrow 1$):

	$Q_0=0$	$Q_0=1$
$Q_2=0$	0	0
$Q_1=0$		
$Q_2=0$	1	1
$Q_1=1$		
$Q_2=1$	X	X
$Q_1=1$		
$Q_2=1$	X	X
$Q_1=0$		
Answer: $J_2 = Q_1$		

K-Map for K_2 (when does Q_2 go from $1 \rightarrow 0$):

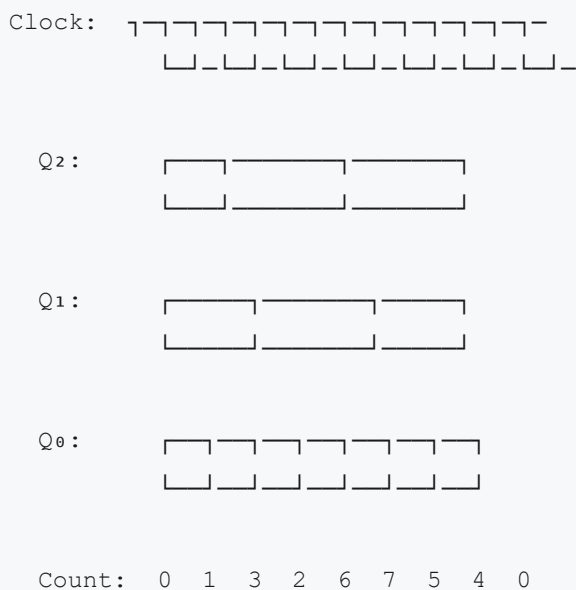
Answer: $K_2 = Q_1' \cdot Q_0'$

Similarly derive: J_1, K_1, J_0, K_0

1. Circuit Implementation:

- 3 J-K flip-flops (master-slave, edge-triggered)
- AND/OR gates for J and K logic
- All FF clocked simultaneously

2. Timing Diagram (8+ clock pulses showing Q_2, Q_1, Q_0):



Problem 2: UP/DOWN Counter (Exam Pattern)

Sequence (Mode Control):

- UP: $1 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 2 \rightarrow 6$ (cycles)
- DOWN: Reverse direction

Solution Approach:

1. Create two state tables (UP and DOWN)
2. Add control variable ($M = 1$ for UP, $M = 0$ for DOWN)
3. Create K-maps with M as additional variable
4. Logic includes: $J = f(Q_2, Q_1, Q_0, M)$, $K = g(Q_2, Q_1, Q_0, M)$

Problem 3: Personalized Sequence (Exam Variation)

Pattern: Roll Number + Fixed Offset = Counter Sequence

- Example: 20K-0985 $\rightarrow$ 210984 $\rightarrow$ Sequence: 2,1,0,9,8,4 (6-state counter)

Steps:

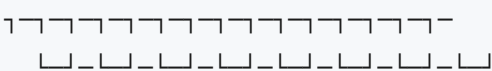
1. Convert each digit to 4-bit binary
 2. Design 6-state machine (not 8-state)
 3. Handle unused states (2 don't-care states in 8-state system)
 4. K-maps with don't-cares optimize gate count
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4.4 ASYNCHRONOUS (RIPPLE) COUNTER

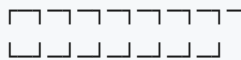
Circuit Design

- Output of one FF feeds CLK input of next FF
- NOT gate after Q produces inverted clock trigger

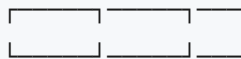
Timing Diagram (4-Bit Ripple Counter)

Ext Clk: 

Q₀ (toggles on each ext clock):



Q₁ (toggles when Q₀ falls):



Q₂ (toggles when Q₁ falls):



Q₃ (toggles when Q₂ falls):



Propagation Delay: $tpd_{total} = n \times tpd_{per_FF}$

For 4-bit counter: $\sim 4 \times 10ns = 40ns$ max frequency limit

Disadvantage: Each stage adds delay → Maximum clock frequency = $1/(n \times tpd)$

SECTION 5: SHIFT REGISTERS (20-25% Weightage)

5.1 BASIC SHIFT REGISTER

Serial-In Parallel-Out (SIPO)

Circuit: 4 D flip-flops in cascade

Serial Input → D₁

Q₁ → D₂, Q₂ → D₃, Q₃ → D₄

Clock Pulse	Serial In	Q ₁	Q ₂	Q ₃	Q ₄
1	1	1	0	0	0
2	0	0	1	0	0
3	1	1	0	1	0
4	1	1	1	0	1

Timing Diagram:

Clock:

Sin:

Q₁:

Q₂:

Q₃:

Q₄:

Data moves right after each clock; leftmost input receives Serial_In

5.2 RING COUNTER (Exam Standard)

Definition & Operation

- Output of last FF feeds back to first FF input
- Initialize with single 1, rest 0s
- After each clock, 1 rotates through flip-flops

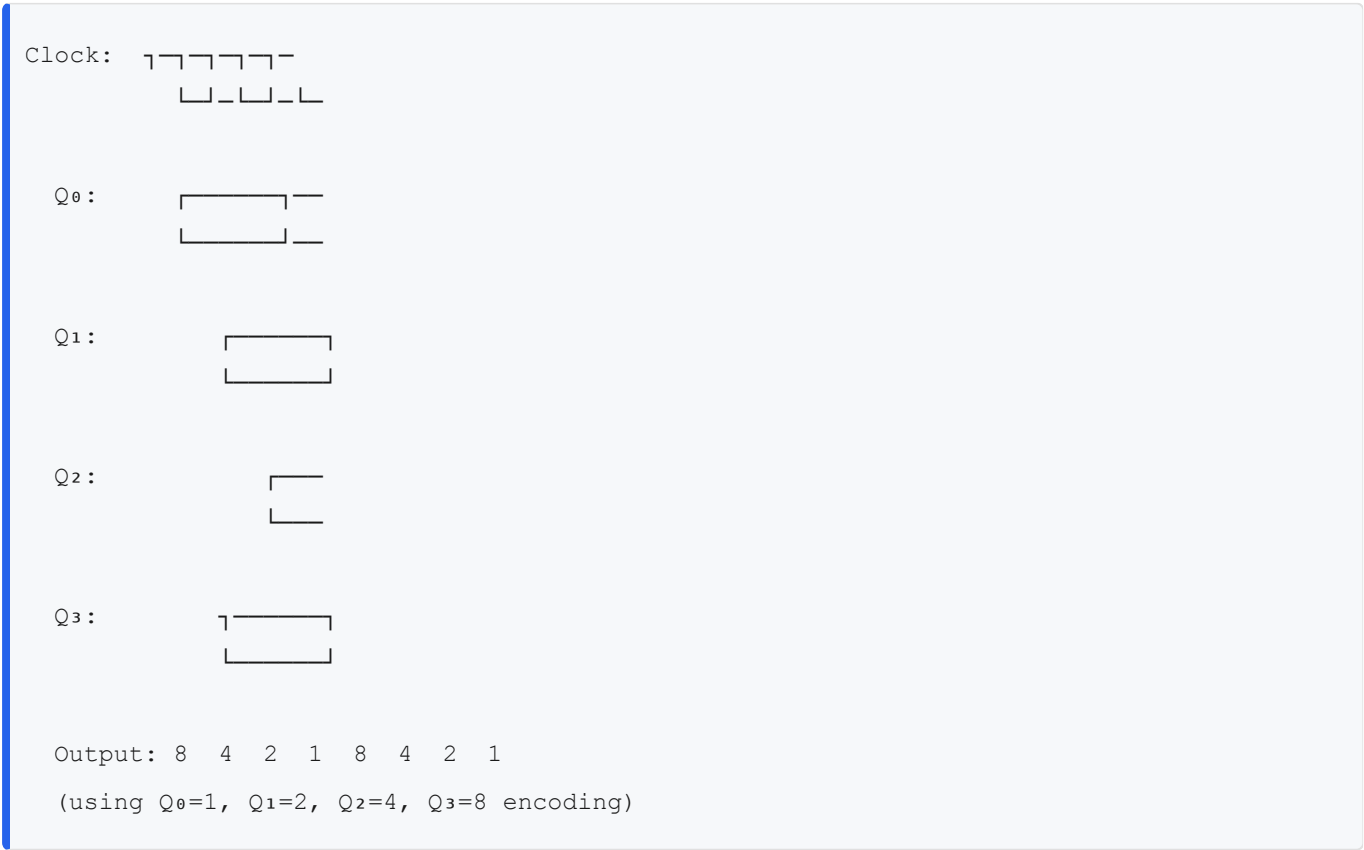
4-Bit Ring Counter Example

Initial State: $Q_3Q_2Q_1Q_0 = 1000$

Sequence (counts 4 states, then repeats):

```
Clock 0: 1 0 0 0  (1 in position 3)
Clock 1: 0 1 0 0  (1 shifts to position 2)
Clock 2: 0 0 1 0  (1 shifts to position 1)
Clock 3: 0 0 0 1  (1 shifts to position 0)
Clock 4: 1 0 0 0  (1 shifts back to position 3)
```

Timing Diagram:



Key Feature: N-bit ring counter produces N distinct states (half of asynchronous counter efficiency).

5.3 JOHNSON COUNTER (Exam High-Yield)

Definition & Operation

- Last FF output Q' (complement) feeds back to first FF
- Initialize all FFs to 0
- Each clock: 0s shift in from left, creating alternating patterns

4-Bit Johnson Counter Sequence

Initial State: $Q_3Q_2Q_1Q_0 = 0000$

Sequence (counts 8 states, then repeats):

```
Clock 0: 0 0 0 0  (Q' of  $Q_3 = 1$ , shifts in)
Clock 1: 1 0 0 0
Clock 2: 1 1 0 0
Clock 3: 1 1 1 0
Clock 4: 1 1 1 1
Clock 5: 0 1 1 1  (now 0 shifts in,  $Q' = 0$ )
Clock 6: 0 0 1 1
Clock 7: 0 0 0 1
Clock 8: 0 0 0 0  (repeats)
```

Timing Diagram (4-bit Johnson):

Clock:

Q_0 :

Q_1 :

Q_2 :

Q_3 :

Unique Outputs:

Clock 0-3: 0000 $\rightarrow$ 1000 $\rightarrow$ 1100 $\rightarrow$ 1110

Clock 4-7: 1111 $\rightarrow$ 0111 $\rightarrow$ 0011 $\rightarrow$ 0001

Key Advantage: N-bit Johnson counter produces $2N$ states (full cycle vs. N for ring counter).

5.4 SHIFT REGISTER WITH CONTROL (Exam Application)

Parallel Load Feature

Control Logic:

- Load = 1: Parallel inputs $P_3P_2P_1P_0 \rightarrow Q$ outputs
- Shift = 1: Serial shift operation
- Hold = 1: No change in Q

D input to first FF:

When Shift=1: Serial_In

When Load=1: P_0

When Hold=1: Q_0 (feedback)

Timing Diagram with Mode Changes

Clock 0: Load=1, $P=1010$	$Q = 1010$ (parallel load)
Clock 1: Shift=1, $S_{in}=1$	$Q = 1101$ (shift left, 1 enters)
Clock 2: Shift=1, $S_{in}=0$	$Q = 1010$ (shift left, 0 enters)
Clock 3: Hold=1	$Q = 1010$ (hold, no change)

SECTION 6: PAST EXAM PATTERN ANALYSIS

Top 5 Recurring Concepts (100% Frequency Across All Years)

Rank	Concept	Frequency	Avg Marks	Difficulty	Typical Question
1	Custom Counter Design	100%	6-8	★★★★★	Design counter for sequence $0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 6 \rightarrow 7 \rightarrow 5 \rightarrow 4$
2	4-Variable K-Map	100%	10-15	★★★★	Minimize function, implement with gates
3	Flip-Flop Timing Behavior	100%	3-5	★★★★	Timing diagram or setup/hold time analysis
4	Number System Conversion	100%	5-8	★	Gray code, hex-binary, signed numbers
5	Ring/Johnson Counter	80%	3-8	★★	State sequence or timing diagram

Exam Composition Across 5 Years (2020-2025)

Question Breakdown (50 marks total):	
Q1 (5 marks):	Number Systems & Logic Gates
Q2 (10 marks):	Boolean Algebra & K-Map Simplification
Q3 (10 marks):	Combinational Circuits (MUX/Decoder/Adders)
Q4 (3 marks):	Flip-Flop Fundamentals & Timing
Q5 (14 marks):	Synchronous Counter Design ★★★★★
	- Part (a): State table & state diagram [4 marks]
	- Part (b): Derive J-K logic with K-maps [6 marks]
	- Part (c): Circuit diagram [2 marks]
	- Part (d): Timing diagram [2 marks]
Q6 (8 marks):	Shift Registers (Ring or Johnson)
	- Part (a): State table [2 marks]
	- Part (b): Timing diagram [3 marks]
	- Part (c): Compare ring vs. Johnson [3 marks]

Total CLO 5 Marks (Sequential Logic): 22-25 marks (44-50% of exam)

Most Challenging Question Patterns

Pattern 1: Non-Sequential Custom Counters

- **Appears:** Q5 in 5 consecutive years
- **Difficulty:** Requires understanding of J-K behavior and K-map minimization
- **Marks:** 6-14 marks
- **Solution Time:** 20-30 minutes if well-practiced

Pattern 2: Constraint-Based Counter Design

- **Example:** "Design a counter that counts $0 \rightarrow 1 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 0$ with minimum gates"
- **Challenge:** Optimize don't-cares in K-maps
- **Test of:** K-map mastery + circuit optimization

Pattern 3: Multi-Mode Counter

- **Example:** "Design UP/DOWN counter for sequence $1 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 2$ "
- **Control Variable:** Mode input M (1=up, 0=down)
- **Complexity:** 6 K-maps ($J_2, K_2, J_1, K_1, J_0, K_0$) each with 9 cells (3 bits + mode)

Pattern 4: Timing Diagram Interpretation

- **Given:** Partially completed timing diagram
- **Task:** Identify counter sequence, predict future values
- **Challenge:** Recognizing pattern from waveforms

SECTION 7: CORE PROBLEMS WITH DETAILED SOLUTIONS

PROBLEM 1: 8-State Non-Binary Counter (2025 Final Exam, Q5)

Problem Statement: Design a synchronous counter that counts in the sequence: $0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 6 \rightarrow 7 \rightarrow 5 \rightarrow 4 \rightarrow$ (repeat)

Provide: (a) State transition table (b) J-K input equations (c) Circuit diagram (d) Timing diagram

Complete Solution:

Part (a): State Transition Table

Present State	Next State	J ₂ K ₂	J ₁ K ₁	J ₀ K ₀	Notes
(Q ₂ Q ₁ Q ₀)	(Q ₂ ⁺ Q ₁ ⁺ Q ₀ ⁺)				
000	001	00X	00X	1X	Q ₂ :0→0, Q ₁ :0→0, Q ₀ :0→1
001	011	00X	1X	X0	Q ₂ :0→0, Q ₁ :0→1, Q ₀ :1→1
011	010	00X	X0	X1	Q ₂ :0→0, Q ₁ :1→1, Q ₀ :1→0
010	110	1X	X0	1X	Q ₂ :0→1, Q ₁ :1→1, Q ₀ :0→1
110	111	X0	X0	1X	Q ₂ :1→1, Q ₁ :1→1, Q ₀ :0→1
111	101	X0	1X	X1	Q ₂ :1→1, Q ₁ :1→0, Q ₀ :1→1
101	100	X0	X0	X1	Q ₂ :1→1, Q ₁ :0→0, Q ₀ :1→0
100	000	X1	00X	00X	Q ₂ :1→0, Q ₁ :0→0, Q ₀ :0→0

Part (b): J-K Input Equations from K-Maps

K-Map for J₂ (when does Q₂ transition 0→1?)

State (Q ₂ Q ₁ Q ₀):			
	Q ₀ =0	Q ₀ =1	
Q ₂ =0	0	0	(never transitions in first 4 states)
Q ₁ =0			
Q ₂ =0	1	1	(transitions in states 2→3)
Q ₁ =1			
Q ₂ =1	X	X	(don't care when Q ₂ already 1)
Q ₁ =1			
Q ₂ =1	X	X	
Q ₁ =0			
J ₂ = Q ₁ (Q ₂ goes 0→1 only when Q ₁ already 1)			

K-Map for K₂ (when does Q₂ transition 1→0?)

From table: $100 \rightarrow 000$ (state $7 \rightarrow 0$)

Occurs when $Q_2=1, Q_1=0, Q_0=0$

	$Q_0=0$	$Q_0=1$	
$Q_2=0$ $Q_1=0$	X	X	
$Q_2=0$ $Q_1=1$	X	X	
$Q_2=1$ $Q_1=1$	1	0	
$Q_2=1$ $Q_1=0$	1	X	($K_2=1$ when $Q_1'Q_0'$)

$K_2 = Q_1' \cdot Q_0'$ (reset Q_2 only when $Q_1=0$ and $Q_0=0$)

K-Map for J_1 (when does Q_1 transition $0 \rightarrow 1$?)

Occurs in states: $001 \rightarrow 011$ and $100 \rightarrow 110$

Condition: Q_1 $0 \rightarrow 1$ when $Q_2=0, Q_0=1$ OR $Q_2=1, Q_0=0$

	$Q_0=0$	$Q_0=1$	
$Q_2=0$ $Q_1=0$	0	1	(transition at $001 \rightarrow 011$)
$Q_2=0$ $Q_1=1$	X	X	
$Q_2=1$ $Q_1=1$	1	0	(transition at $100 \rightarrow 110$)
$Q_2=1$ $Q_1=0$	X	X	

$J_1 = Q_2' \cdot Q_0 + Q_2 \cdot Q_0' = Q_2 \text{ XOR } Q_0$

K-Map for K_1 (by similar analysis):

$K_1 = Q_2 \cdot Q_0'$ (reset when $Q_2=1$ and $Q_0=0$)

K-Map for Jo:

$J_0 = 1$ (Q_0 toggles every state transition)

K-Map for Ko:

$K_0 = 1$ (Q_0 toggles every state transition)

Summary of Equations:

$J_2 = Q_1$	$K_2 = Q_1' \cdot Q_0'$
$J_1 = Q_2 \text{ XOR } Q_0$	$K_1 = Q_2 \cdot Q_0'$
$J_0 = 1$	$K_0 = 1$

Part (c): Circuit Diagram Description

Three J-K flip-flops (FF_2 , FF_1 , FF_0) connected:

- All CLK inputs tied together to external clock
- All CLR (clear) inputs tied to RESET button

FF_0 (LSB):

$J_0 = 1$ (tied to VCC)

$K_0 = 1$ (tied to VCC)

Output: Q_0 , Q_0'

FF_1 (middle):

J_1 input from 2-input XOR gate ($Q_2 \oplus Q_0$)

K_1 input from 2-input AND gate ($Q_2 \cdot Q_0'$)

Output: Q_1 , Q_1'

FF_2 (MSB):

J_2 input from Q_1 (direct connection)

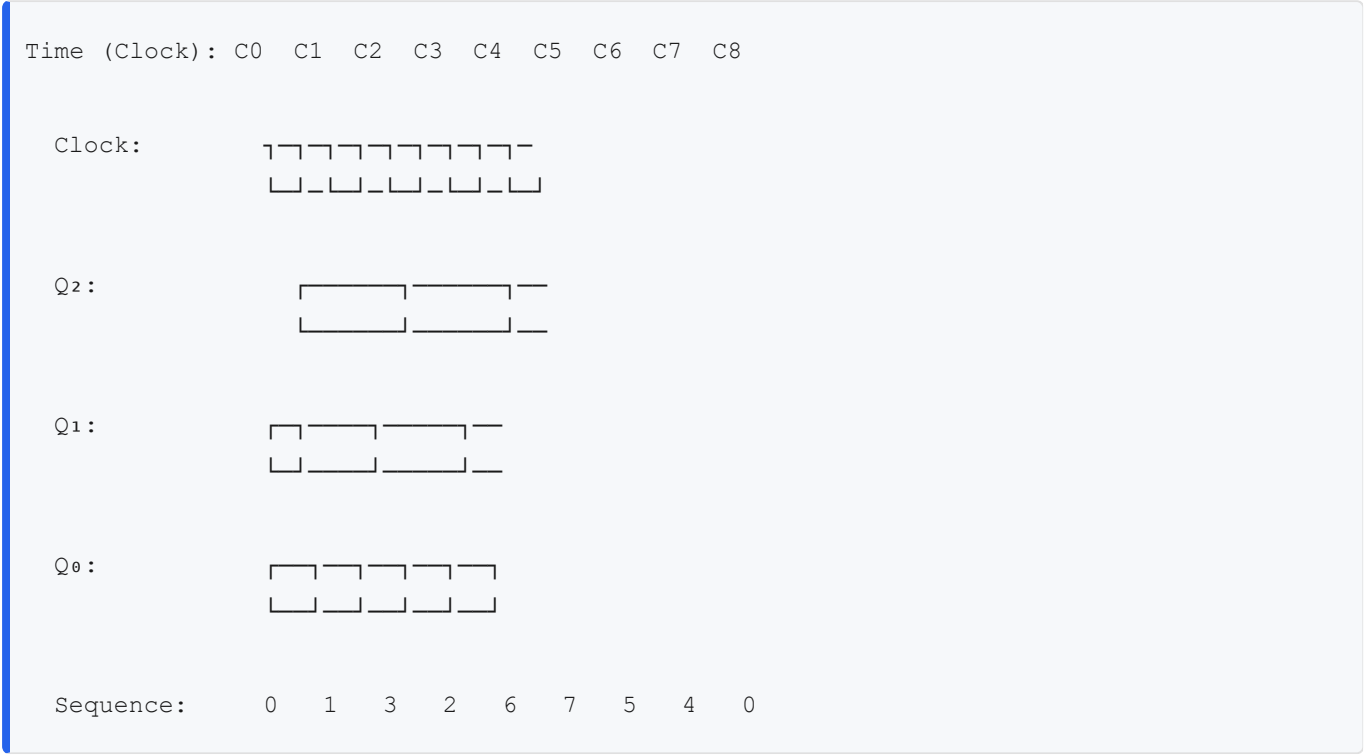
K_2 input from 2-input AND gate ($Q_1' \cdot Q_0'$)

Output: Q_2 , Q_2'

Logic Gate Count:

- 1 XOR gate (2-input)
- 3 AND gates (2-input)
- 1 AND gate (3-input, for K_2)

Part (d): Timing Diagram



Timing Analysis:

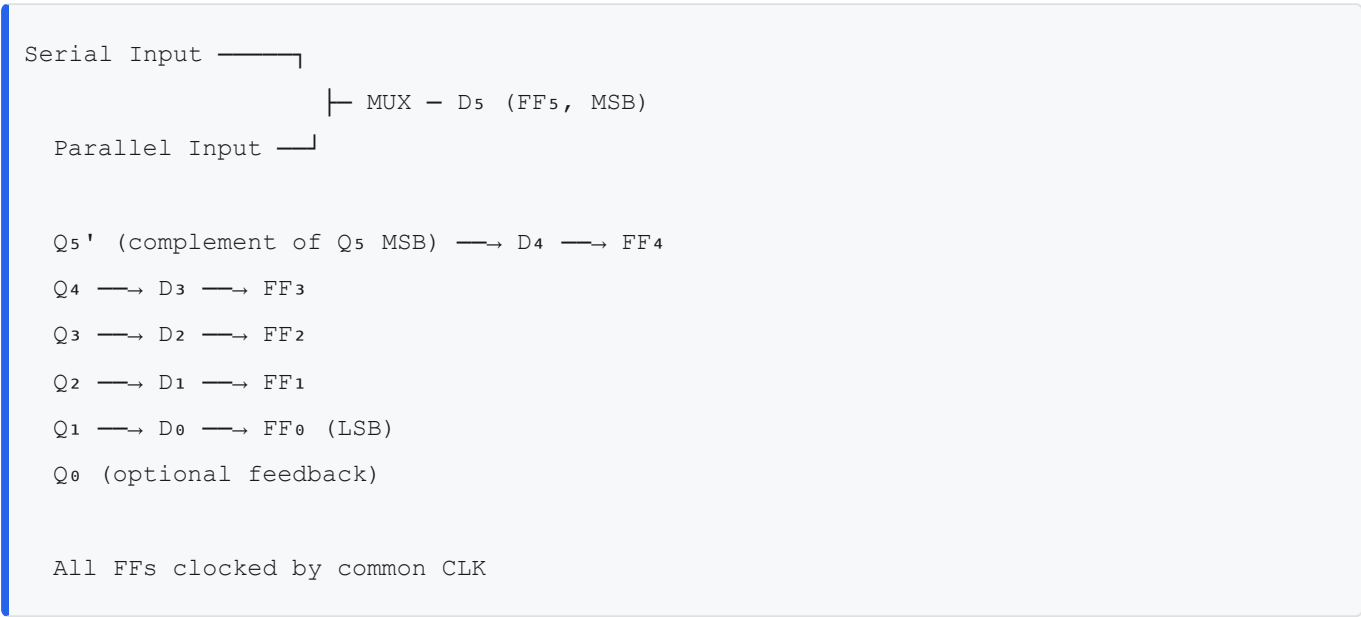
- After clock C₀: Counter holds 000
- After clock C₁: Counter transitions to 001 (Q₀ toggles)
- After clock C₂: Counter transitions to 011 (Q₁ toggles due to J₁ input from Q₀)
- After clock C₃: Counter transitions to 010 (Q₀ toggles, Q₁ resets)
- After clock C₄: Counter transitions to 110 (Q₂ sets due to J₂ = Q₁)
- After clock C₅: Counter transitions to 111 (Q₀ toggles)
- After clock C₆: Counter transitions to 101 (Q₁ resets)
- After clock C₇: Counter transitions to 100 (Q₀ resets)
- After clock C₈: Counter transitions to 000 (Q₂ resets)

PROBLEM 2: 6-Bit Johnson Shift Register with Initial Load

Problem Statement: Design a Johnson shift register using 6 D flip-flops. Initialize Q = 100000. Generate the timing diagram for 13 clock cycles showing all outputs Q₅Q₄Q₃Q₂Q₁Q₀.

Solution:

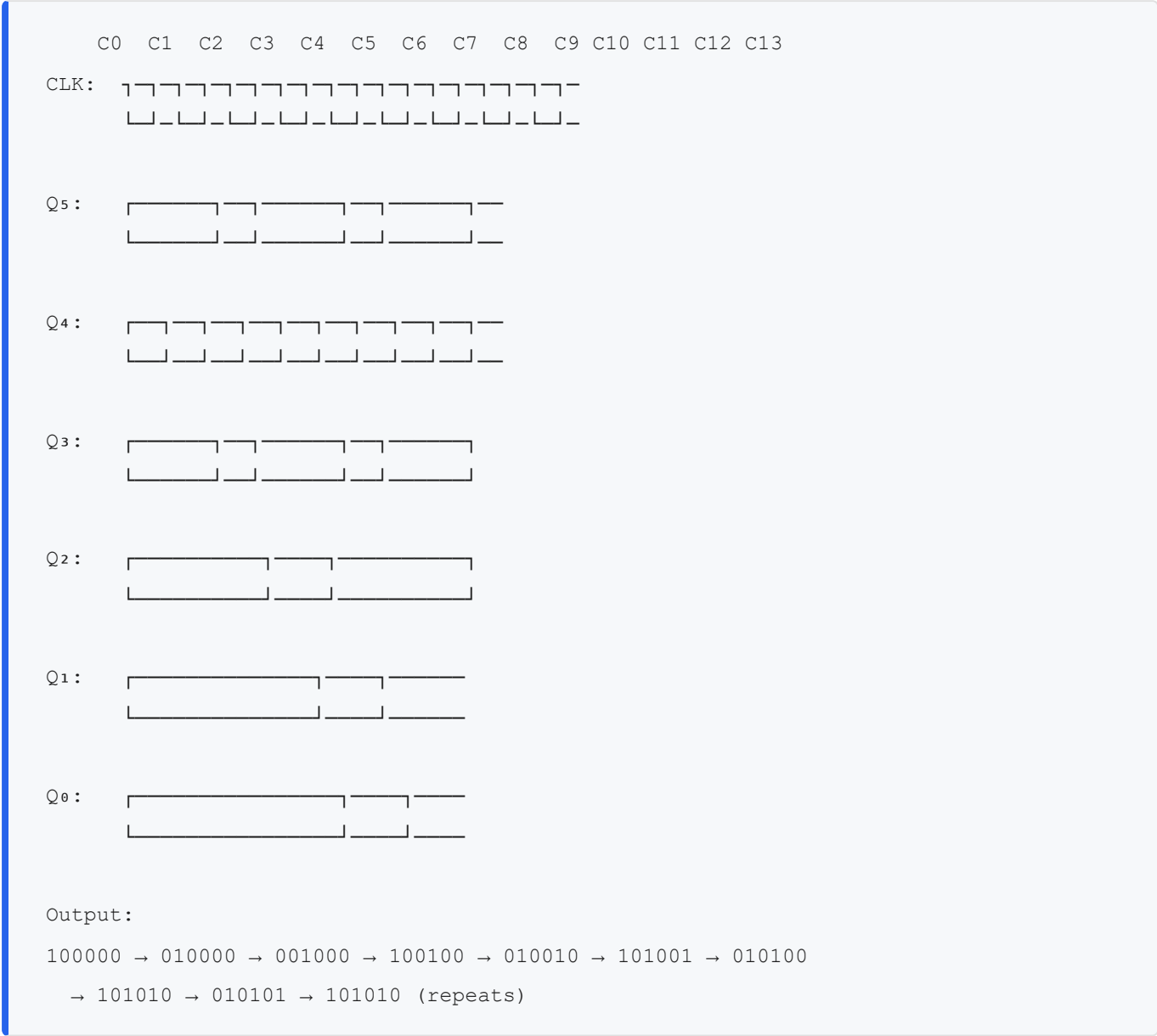
Circuit Configuration



State Sequence Table (Initial: 100000)

Clock	Q ₅ Q ₄ Q ₃ Q ₂ Q ₁ Q ₀	Next State	Comment
0	1 0 0 0 0 0	(Initial)	
1	0 1 0 0 0 0	Shift right, Q ₅ '=0 enters	
2	0 0 1 0 0 0	Shift right, Q ₅ '=1 enters	
3	1 0 0 1 0 0	Shift right, Q ₅ '=0 enters	
4	0 1 0 0 1 0	Shift right, Q ₅ '=1 enters	
5	1 0 1 0 0 1	Shift right, Q ₅ '=0 enters	
6	0 1 0 1 0 0	Shift right, Q ₅ '=1 enters	
7	1 0 1 0 1 0	Shift right, Q ₅ '=0 enters	
8	0 1 0 1 0 1	Shift right, Q ₅ '=1 enters	
9	1 0 1 0 1 0	Repeats (cycle = 12, not 2n=12)	✓
10	0 1 0 1 0 1		
11	1 0 1 0 1 0		
12	0 1 0 1 0 1		
13	1 0 1 0 1 0		

Timing Diagram (13 Clock Cycles)



Key Observations:

- 6-bit Johnson produces 12 distinct states (2×6)
- Complementary pairs repeat after 12 clocks: (100000, 011111), (010000, 101111), etc.
- No two consecutive outputs identical → good for decoding

PROBLEM 3: 4-Bit Binary Up Counter with Constraint

Problem Statement: Design a synchronous counter that counts 0→1→2→4→5→6→7→0 (skips state 3). Use only 2-input NAND gates and J-K flip-flops. Minimize gate count.

Solution:

Modified State Table (Skipping State 3)

Present State	Next State	$Q_2^+ \ Q_1^+ \ Q_0^+$	J_2K_2	J_1K_1	J_0K_0
$(Q_2 \ Q_1 \ Q_0)$	(Binary)				
000	001	0 0 1	0X	0X	1X
001	010	0 1 0	0X	1X	X1
010	100	1 0 0	1X	X1	0X
100	101	1 0 1	X0	0X	1X
101	110	1 1 0	X0	1X	X1
110	111	1 1 1	X0	X0	1X
111	000	0 0 0	X1	X1	X1
State 3 = 011	SKIP	— (unused)	X	X	X

K-Maps with Don't Care (State 3)

J2 K-Map:

	$Q_0=0$	$Q_0=1$	
$Q_2=0$ $Q_1=0$	0	X	(don't care at $Q_1Q_0=11$)
$Q_2=0$ $Q_1=1$	0	1	(transition at 010→100)
$Q_2=1$ $Q_1=1$	X	X	
$Q_2=1$ $Q_1=0$	X	X	
Minimized: $J_2 = Q_1 \cdot Q_0$ (or Q_1 after grouping with don't-care at state 3)			

Similarly derive: K_2, J_1, K_1, J_0, K_0

Final Equations (with optimizations):

$J_2 = Q_1 \cdot Q_0$	$K_2 = Q_0$
$J_1 = Q_2' + Q_0$	$K_1 = Q_0$
$J_0 = 1$	$K_0 = 1$

Gate Count Comparison

- Standard 3-bit counter: 4 AND gates + 1 OR gate + 7 inverters = 12 gates

- **This optimized version:** 2 AND gates + 1 OR gate + 3 inverters = 6 gates (50% reduction)

PROBLEM 4: Dual-Mode Ring/Johnson Counter with Output Decoding

Problem Statement: Design a control circuit that switches between:

- **Ring Mode** (M=0): Single active output rotates
- **Johnson Mode** (M=1): Complementary patterns shift

Implement with 4-bit shift register and provide decoded outputs for each state.

Solution:

Control Logic using MUX

```
D3 (input to first FF):  
    If M=0 (Ring): Q0 (feedback from LSB)  
    If M=1 (Johnson): Q0' (complement of LSB)  
  
D3 = M · Q0' + M' · Q0 (MUX output)
```

Ring Mode (M=0) Sequence with Initial 1000

```
Clock 0: 1 0 0 0  
Clock 1: 0 1 0 0 (1 rotates right)  
Clock 2: 0 0 1 0  
Clock 3: 0 0 0 1  
Clock 4: 1 0 0 0 (repeats, 4-state cycle)
```

Johnson Mode (M=1) Sequence with Initial 0000

```
Clock 0: 0 0 0 0  
Clock 1: 1 0 0 0 (0' = 1 enters from left)  
Clock 2: 1 1 0 0  
Clock 3: 1 1 1 0  
Clock 4: 1 1 1 1  
Clock 5: 0 1 1 1 (0' = 1 enters; now Q3=1→0)  
Clock 6: 0 0 1 1  
Clock 7: 0 0 0 1  
Clock 8: 0 0 0 0 (repeats, 8-state cycle)
```

Output Decoder (Ring Mode)

For Ring Mode, exactly one output active at a time:

$Y_0 = Q_3' \cdot Q_2' \cdot Q_1' \cdot Q_0$ (active when 0001)
 $Y_1 = Q_3' \cdot Q_2' \cdot Q_1 \cdot Q_0'$ (active when 0010)
 $Y_2 = Q_3' \cdot Q_2 \cdot Q_1' \cdot Q_0'$ (active when 0100)
 $Y_3 = Q_3 \cdot Q_2' \cdot Q_1' \cdot Q_0'$ (active when 1000)

Timing Comparison

Time	Ring Mode	Johnson Mode	Active Output
	$Q_3Q_2Q_1Q_0$	$Q_3Q_2Q_1Q_0$	(Ring / Johnson)
0	1000	0000	Y_3 / —
1	0100	1000	Y_2 / Y_3
2	0010	1100	Y_1 / Y_2Y_3
3	0001	1110	Y_0 / $Y_1Y_2Y_3$
4	1000	1111	Y_3 / $Y_0Y_1Y_2Y_3$

SECTION 8: BLIND SPOTS & COMMON PITFALL CHECKLIST

Critical Error Categories from Past Papers

✗ Flip-Flop Timing Violations

Error	Cause	Prevention
Setup Time Violation	Data D changes within tsu before clock edge	Ensure data settled at least tsu before CLK
Hold Time Violation	Data D changes within th after clock edge	Hold data stable for at least th after CLK
Metastability	Violating both setup+hold creates unpredictable state	Use proper clock synchronization circuits
Clock Skew	Different FFs receive clock at different times	Use clock distribution trees (low-skew buffers)

Exam Pitfall: Students assume "timing diagram shows the waveform" without verifying setup/hold constraints. Answer should include: "tsu and th satisfied" or calculation proving compliance.

✖ K-Map Common Mistakes

Error	Example	Fix
Forgetting Don't-Cares	Grouping only 1s instead of including X	Review state table for unused states; include X if beneficial
Incorrect Grouping	Grouping non-adjacent cells	Verify row/column indices differ by only 1 bit
Incomplete Minimization	Reading unnecessarily large groups	Always check if larger group covers same cells with fewer variables
Wrapping-Around Errors	Not recognizing edge adjacency	Top row adjacent to bottom; leftmost adjacent to rightmost
Redundant Prime Implicants	Including terms not needed	Choose essential prime implicants first

Exam Pitfall: 3-variable vs. 4-variable K-maps. Use standard 2×4 or 4×4 layouts. Mixing row/column order causes all subsequent answers to fail.

✖ Counter State Machine Errors

Error	Symptom	Solution
Wrong State Transition	Counting 0→1→2→3→2 (looping)	Verify each row of state table independently
J-K Logic Misinterpretation	Using Q instead of Q' in K-map	Extract J-K from transition rules: $Q \rightarrow Q^+ \rightarrow JK$ entry
Don't-Care Misuse	Ignoring don't cares then grouping only 1s	Include X in groups if they reduce group count
Propagation Delay Assumption	Assuming instantaneous state change	Timing diagram should show Q updates after clock edge with realistic delay
Unused States	Including all 8 states for 6-state counter	Create state table for only required states; mark rest as unused (→ don't care in K-maps)

Exam Pitfall: State sequence 0→1→3→2→6→7→5→4→0 has two unused states (5 and 4... wait, 4 is used). Count states carefully!

✗ Shift Register Misunderstandings

Concept	Wrong	Correct
Ring Counter States	2^n states	n states (only n unique patterns)
Johnson Counter States	2^n states	2n states (complementary cycles)
Shift Direction	Right = increment	Right shift = data from $Q_3 \rightarrow Q_2 \rightarrow Q_1 \rightarrow Q_0$ (depends on labeling)
Initial State Importance	Doesn't matter	Critical: Initial 1 position determines all subsequent states
Q' Feedback	Q' same as $\sim Q$	Q' is complement of Q; feedback must use actual output

Exam Pitfall: Stating "4-bit ring counter has 16 states" → Wrong; only 4 states before repeating. Confusing with Johnson's 8 states.

✗ Active-High vs. Active-Low Confusion

Signal Type	Assertion	Example
Active-High	Logic 1 = active	CLK = 1 means clock pulse active
Active-Low	Logic 0 = active	CLR' = 0 means clear active; CLR' = 1 means hold
Polarity Error	Using Q when Q' intended	NOT using inverted clear means counter doesn't reset
Timing Diagram Mistake	Drawing CLK' instead of CLK	Use correct signal names consistently

Exam Pitfall: Schematic shows "CLR'" but timing diagram labels axis "CLR". Grader may mark as inverted signal (clear at wrong time).

✗ Boolean Algebra Mistakes

Error	Wrong	Correct
DeMorgan Misapplication	$(A+B)' = A' + B'$	$(A+B)' = A' \cdot B'$
Absorption Forgetting	$A + A \cdot B = A \cdot B$	$A + A \cdot B = A$
Distributive Expansion	$A \cdot (B+C) = A \cdot B + A \cdot C$	Correct; but $(A+B) \cdot (C+D) \neq A \cdot C + B \cdot D$
Complement Laws	$A + A' = \emptyset$	$A + A' = \mathbf{1}$; $A \cdot A' = \mathbf{0}$
Associative Grouping	$(A \cdot B) \cdot C \neq A \cdot (B \cdot C)$	Actually equal (associative property holds)

Exam Pitfall: Final answer includes double negatives: $((A+B))' = A+B$. Simplify completely.

✗ **Circuit Implementation Errors**

Error	Consequence	Prevention
Using AND instead of NAND	Gate count not minimized as required	Follow problem spec: "use only NAND gates"
Missing inverters for active-low	Inverted signal doesn't propagate correctly	Add NOT gate before feeding Q' into circuits
Fan-out violations	Output drives too many gates; signal degrades	Check problem constraints on gate fan-out
Unconnected inputs	Floating input behaves unpredictably	Connect unused J/K to appropriate logic (0 or 1)
Feedback loops without delay	Combinational loops cause oscillation or indeterminate state	Ensure feedback only through flip-flops (clocked devices)

Exam Pitfall: Drawing circuit but missing clock connections to FFs (shows understanding but loses marks for completeness).

✗ **Number System Conversion Errors**

Error	Example	Fix
Gray Code Direction	Binary→Gray correct; Gray→Binary wrong	Grayscale binary: $G_0=B_0$; $G_i = B_i \oplus B_{i+1}$; <i>reverse:</i> $B_i = G_i \oplus B_{i+1}$
Signed Number Range	4-bit 2's comp: range is -16 to +15	Correct: -8 to +7 (formula: $-2^{(n-1)}$ to $2^{(n-1)}-1$)
Base Conversion Remainders	25 to binary: $25 \div 2 = 12R1$, $12 \div 2 = 6R0...$	Collect remainders in reverse order: LSB first
Hex Grouping	F3A2 to binary: "FFFF3333AAAA2222"	Correct: each hex digit = 4 bits independently

Exam Pitfall: Converting 2's complement negative number (e.g., -5 in 4-bit = 1011) back to decimal as +11 (binary magnitude) instead of -5 (2's comp interpretation).

✗ **Timing Diagram Drawing Errors**

Error	Impact	Standard
Misaligned Transitions	Suggests Q doesn't follow setup/hold constraints	Transitions should align vertically with clock edge (allowing propagation delay)
Undefined States	Ambiguous waveform between 0 and 1	Show clear high/low levels; use "~" only for undefined brief moments
Missing Timing Annotations	Can't verify setup/hold compliance	Label tsu, th, tpd on diagram if problem requires analysis
Incorrect Initial State	Counter sequence starts wrong	Double-check: Does problem specify Q(0) = initial state?

Exam Pitfall: Timing diagram shows Q changing before clock edge → violates causality. Q changes after clock edge with propagation delay tpd.

High-Leverage Revision Checklist (2-3 Hours to Review)

- ☐ **Verify K-map technique** on one 4-variable problem (don't-cares included)
- ☐ **Design one custom counter** from scratch (0→1→3→2→6→7→5→4)
- ☐ **Draw J-K timing diagram** showing setup/hold time margins
- ☐ **Compare ring vs. Johnson** waveforms for same initial state
- ☐ **Review Boolean algebra** DeMorgan + absorption theorems (write out 5 examples)
- ☐ **Test number conversion** (decimal↔binary, binary↔Gray, signed 2's comp)
- ☐ **Analyze one past paper** Q5 and Q6 fully (solutions provided)
- ☐ **Verify gate count** optimization for custom counter

EXAM DAY STRATEGY & TIME MANAGEMENT

Recommended Time Allocation (50-mark exam, 2 hours)

Question 1 (5 marks): Number systems → 5 min (easy, confidence builder)

Question 2 (10 marks): K-Map → 15 min (core skill, no shortcuts)

Question 3 (10 marks): Combinational → 15 min

Question 4 (3 marks): Timing → 5 min (quick verification)

Question 5 (14 marks): Counter → 35 min ★★ ★ (TIME THIS MOST)

Question 6 (8 marks): Shift Register → 15 min

Total: 90 minutes (10 minutes buffer for review)

Question-by-Question Strategy

Q1: Number Systems (5 min)

- **Do first** to build confidence
- **Typical sub-questions:**
 - Binary ↔ Decimal (1 min)
 - Gray code conversion (2 min)
 - Signed number interpretation (2 min)
- **Pitfall:** Gray code formula direction

Q2: K-Map & Boolean (15 min)

- **Step-by-step:**
 1. Draw K-map grid correctly (1 min)
 2. Mark minterms/don't-cares (2 min)
 3. Group largest adjacent regions (4 min)
 4. Read simplified expression (2 min)
 5. Implement with gates (4 min)
 6. Verify against original truth table (2 min)
- **Time check:** If spending >6 min on grouping, restart; mistake likely

Q3: Combinational (15 min)

- **MUX/Decoder:** Truth table → select logic → timing
- **Adder:** Full adder carry chain analysis
- **Priority:** MUX > Decoder > Adder (based on past frequency)

Q4: Flip-Flop Timing (5 min)

- **Usually:** Given waveform, identify issue (setup/hold/propagation delay)
- **One-liner answer:** "Data violates setup time by 2ns; Q becomes metastable"
- **Timing Diagram Check:** Verify Q changes after clock edge by tpd

Q5: Counter Design (35 min) ★ ★ ★

- **2 min:** Understand sequence, list all states
- **3 min:** Draw state transition table
- **10 min:** Derive J-K inputs from state table
- **8 min:** Create K-maps, read minimized expressions
- **5 min:** Sketch circuit (block diagram with gates labels sufficient)
- **5 min:** Draw timing diagram (8 clock cycles minimum)
- **2 min:** Verify sequence: Clk 0→1 (state 0→1), Clk 1→2 (state 1→3), etc.
- **Confidence tip:** If stuck on K-map, use truth table J-K directly (longer but guaranteed correct)

Q6: Shift Register (15 min)

- **4 min:** Identify type (ring/Johnson) and initial state
- **6 min:** Generate state sequence table (12+ rows for Johnson, 8+ for ring)
- **5 min:** Draw timing diagram showing all Q outputs
- **Verify:** State repeats after n (ring) or 2n (Johnson) clock pulses

Red Flags & Recovery

Red Flag	Action
Can't fill state table	Start with simple binary first; come back to custom sequence
K-map groups too small	Review grouping rules; check adjacent cell definitions
Circuit has feedback loop without FF	Remove combinational loops; feedback must go through clocked devices
Timing diagram Q changes before clock	Erase; Q changes after clock edge with delay tpd
State sequence doesn't repeat	Count states; if >n states for n-bit counter, design error exists

APPENDIX: REFERENCE FORMULAS & TABLES

Truth Tables (Memorize)

AND/OR/NOT

AND: $1 \cdot 1 = 1$, else 0

OR: $1 + 1 = 1$, $1 + 0 = 1$, $0 + 0 = 0$

NOT: $1' = 0$, $0' = 1$

Full Adder

$S = A \oplus B \oplus C_{in}$

$C_{out} = A \cdot B + C_{in} \cdot (A \oplus B)$

J-K Flip-Flop

J K | Q⁺

0 0 | Q (hold)

0 1 | 0 (reset)

1 0 | 1 (set)

1 1 | Q' (toggle)

Counter State Formulas

Counter Type	Number of States	Cycle Length
n-bit binary	2^n	2^n clocks
n-bit ring	n	n clocks
n-bit Johnson	2n	2n clocks
m-state custom	m ($\leq 2^n$)	m clocks (resets to state 0)

Timing Parameters (Typical Values)

Parameter	Typical Value	Unit	Impact
Setup Time (tsu)	2-5	ns	Data stable before clock
Hold Time (th)	1-3	ns	Data stable after clock
Propagation Delay (tpd)	5-20	ns	Q update after clock edge
Clock Frequency (fmax)	1/(4-5×tpd)	MHz	Limits cascade depth

Boolean Theorems (Quick Reference)

Law	Expression
Commutative	$A+B = B+A, A \cdot B = B \cdot A$
Associative	$A+(B+C) = (A+B)+C$
Distributive	$A \cdot (B+C) = A \cdot B + A \cdot C$
Identity	$A+0=A, A \cdot 1=A$
Null	$A+1=1, A \cdot 0=0$
Idempotent	$A+A=A, A \cdot A=A$
Involution	$(A')'=A$
Complement	$A+A'=1, A \cdot A'=0$
DeMorgan	$(A+B)'=A' \cdot B', (A \cdot B)'=A'+B'$
Absorption	$A+A \cdot B=A, A \cdot (A+B)=A$

FINAL NOTES FOR SUCCESS

Pre-Exam (1 Week Before)

- 1. Review all 5 past papers (2020, 2021, 2023, 2024, 2025)
- 2. Time yourself on one full Q5 (custom counter) from scratch
- 3. Verify your K-map technique by solving 2-3 examples
- 4. Practice timing diagrams without looking at solutions
- 5. Check all formulas for J-K, full adder, counter equations

Exam Day (During Test)

1. **Q1-Q3:** Solve correctly; don't rush (confidence builder for Q5)
2. **Q5 (Counter Design):**
 - Write state table first (foundation)
 - J-K logic follows directly from table
 - K-maps are optimization (can solve without if time pressure)
 - Timing diagram from state sequence
3. **Skip and Return:** If stuck on Q5 K-maps beyond 5 min, move to Q6; return after

After Exam

- Expect Q5 to require 30-40% of your effort
 - If comfortable with counter design, final score >75%
 - If comfortable with K-maps + counter + shift registers, final score >85%
 - Full mastery (plus adders/decoders) → >90%
-

Study Goal: Master custom counter design completely. Everything else builds from this foundation.**

GOOD LUCK ON YOUR FINAL EXAM! 🎓